

Fine-tuning language model embeddings to reveal pathology domain knowledge

Ceca Krašniković
Alumni Graz University of Technology, Austria

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My research background

PhD research @ Graz University of Technology:

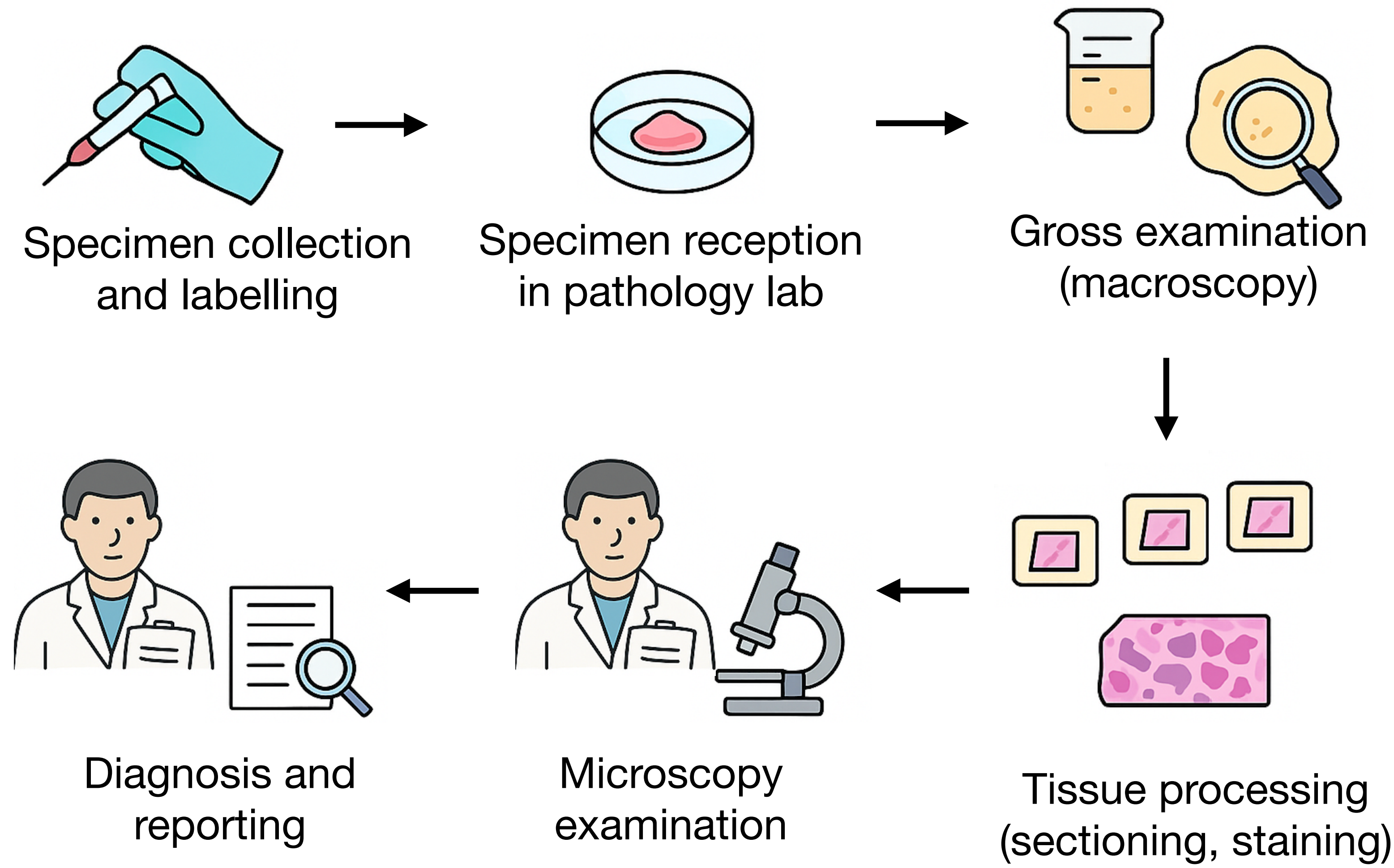
- Deep learning & Computational Neuroscience & Brain-inspired computing
 - Models to understand mechanisms of the brain
 - Brain as inspiration for models that manipulate symbols and use rules
 - Brain-inspired, in-hardware model training
- **Deep learning model training/optimization**
- **Model reverse-engineering:** analysis of model (neural network's) activity to understand the computations

Postdoc @ Medical University of Graz:

- Data management
- Tools for supporting research data access
- **Large Language Models (LLMs) for pathology text reports**
 - Data annotations
 - “Virtual pathology assistant” (e.g., for medical knowledge transfer and learning)

Introduction

Pathology lab workflow



Introduction

- Digital Pathology vs. Computational Pathology [1, 2]
 - Digital pathology:
scanning tissue-slides into high-resolution images - **Whole-Slide-Images** (WSI);
software for more-efficient workflow.
 - Computational pathology: use of advanced algorithms.
 - Central to both: WSIs, accompanied by text reports.
- Artificial Intelligence (AI) algorithms:
 - Require large data sets
 - Image-based, text-based; multi-modal input models...

[1] Jahn, S. W., Plass, M., & Moinfar, F. (2020). Digital pathology: advantages, limitations and emerging perspectives. Journal of clinical medicine, 9(11), 3697.

[2] Shmatko, A., Ghaffari Laleh, N., Gerstung, M., & Kather, J. N. (2022). Artificial intelligence in histopathology: enhancing cancer research and clinical oncology. Nature cancer, 3(9), 1026-1038. ⁴



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Fine-tuning language model embeddings to reveal domain knowledge: An explainable artificial intelligence perspective on medical decision making

Ceca Krašniković^a, Robert Harb^{a,b}, Markus Plass^{a,*}, Wael Al Zoughbi^a, Andreas Holzinger^{a,c,d}, Heimo Müller^a

^a Diagnostic and Research Institute of Pathology, Medical University of Graz, Neue Stiftingtalstraße 6, Graz, 8010, Austria

^b Institute for Computer Graphics and Vision, Graz University of Technology, Inffeldgasse 16, Graz, 8010, Austria

^c Human-Centered AI Lab, Institute for Forest Engineering, University of Natural Resources and Life Sciences Vienna, Peter Jordan Straße 82, Vienna, 1190, Austria

^d Institute of Medical Informatics, Statistics and Documentation, Medical University of Graz, Auenbruggerplatz 2, Graz, 8036, Austria



Motivation

- In-house data set (~1.3 million reports from ~600 thousand patients, collected over 23 years)
- Apply common large model training practices, check usefulness and benefits
- Model reverse-engineering:
explore contextual representations/embeddings & meeting interpretability needs
- Potential benefits for digital and computational pathology:
 - Increase workflow efficiency (quality checks, smart annotations, irregularity alerts)
 - Accelerate specialist training
 - Enhance annotations and enable advanced multi-modal algorithms
 - Improve data search and filtering for research purposes

Introduction:

- Pathology reports: rich source of information about cancer characteristics
 - LLMs, BERT model
 - Fine-tuning
 - Domain-adaptive pre-training + fine-tuning [3]
 - Applications in pathology/medicine:
 - Data privacy, ethical concerns
 - Opening the “black-box”: explainability and interpretability
- ➔ Analysis of contextual representations / embedding (e.g., similarity and separability of representations [4, 5], organisation of knowledge in the embedding space)

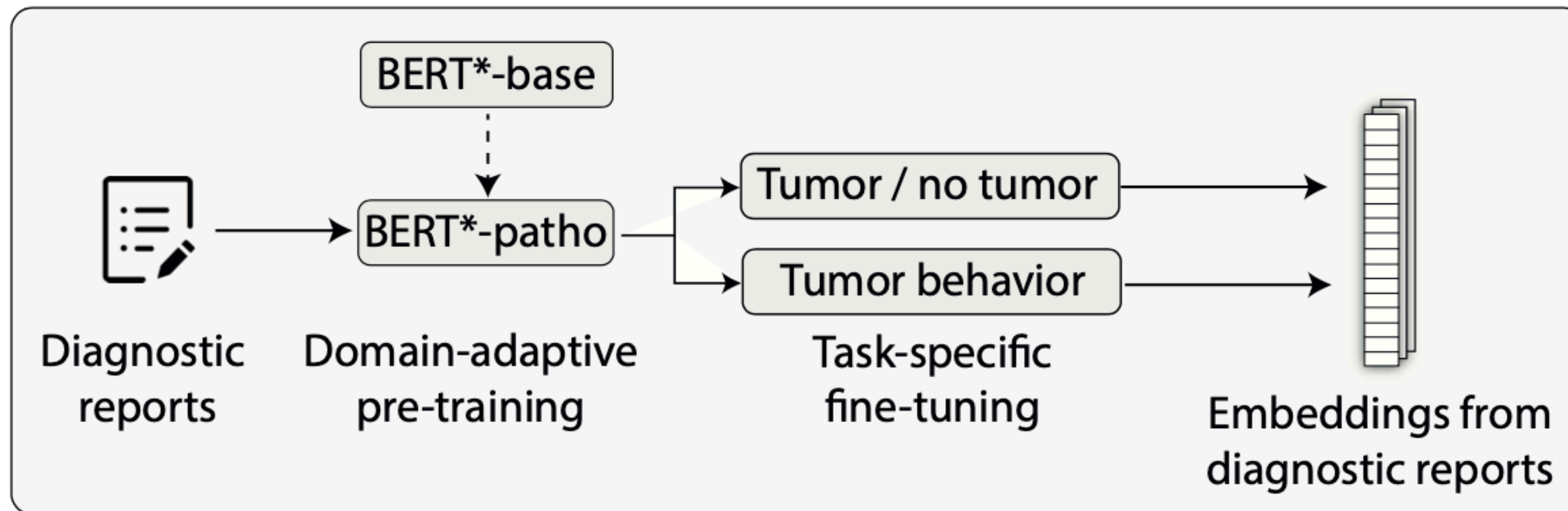
[3] Gururangan, S., Marasović, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., & Smith, N. A. (2020). Don't stop pretraining: Adapt language models to domains and tasks. arXiv preprint arXiv:2004.10964.

[4] Zhou, Y., & Srikumar, V. (2021). A closer look at how fine-tuning changes BERT. arXiv preprint arXiv:2106.14282.

[5] Durrani, N., Sajjad, H., Dalvi, F., & Belinkov, Y. (2020). Analyzing individual neurons in pre-trained language models. arXiv preprint arXiv:2010.02695.

Methods

- **Pathology-specific tasks for fine-tuning:**
 - Tumor presence: Tumor / No tumor
 - Tumor behavior: Benign / Uncertain / In situ / Malignant Primary / Malignant metastatic



Embedding computation

*BERT: Bidirectional Encoder Representations from Transformers

Results

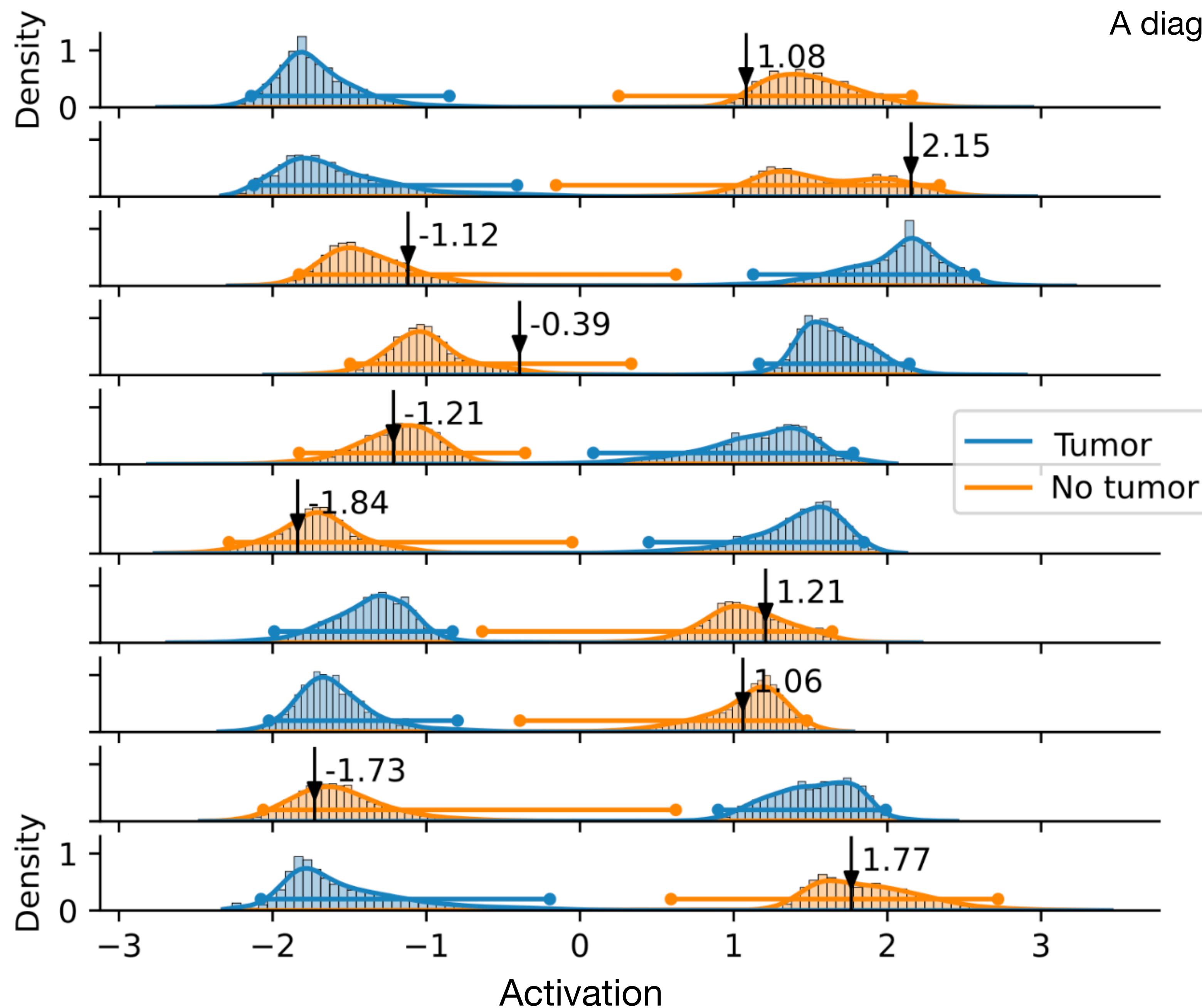
Table 1

Performance results of different fine-tuned models on two pathology-specific tasks.

Task	Fine-tuned model	Training loss	Evaluation loss	Accuracy
Tumor - No tumor	BERT-base	0.108 ± 0.001	0.072 ± 0.008	0.983 ± 0.001
	BERT-patho	0.104 ± 0.001	0.073 ± 0.005	0.982 ± 0.001
Tumor behavior	BERT-base	0.06 ± 0.002	0.042 ± 0.010	0.975 ± 0.006
	BERT-patho	0.06 ± 0.002	0.043 ± 0.006	0.974 ± 0.003

- Both models achieve good results, no significant difference
 - Proceed with embedding analyses to search for the differences
 - Effects of fine-tuning:
 - Similarity analysis: Fine-tuned models produce representations that maximize similarity within the same class while minimizing similarity to representations from different classes.
 - Separability analysis: Linear separability of representations gets enhanced, surprisingly even within single dimensions
- generate visualisations

Single dimensions of the embedding



A diagnosis represented by the vector:

[1.08,

2.15,

-1.12,

-0.39,

-1.21,

-1.84,

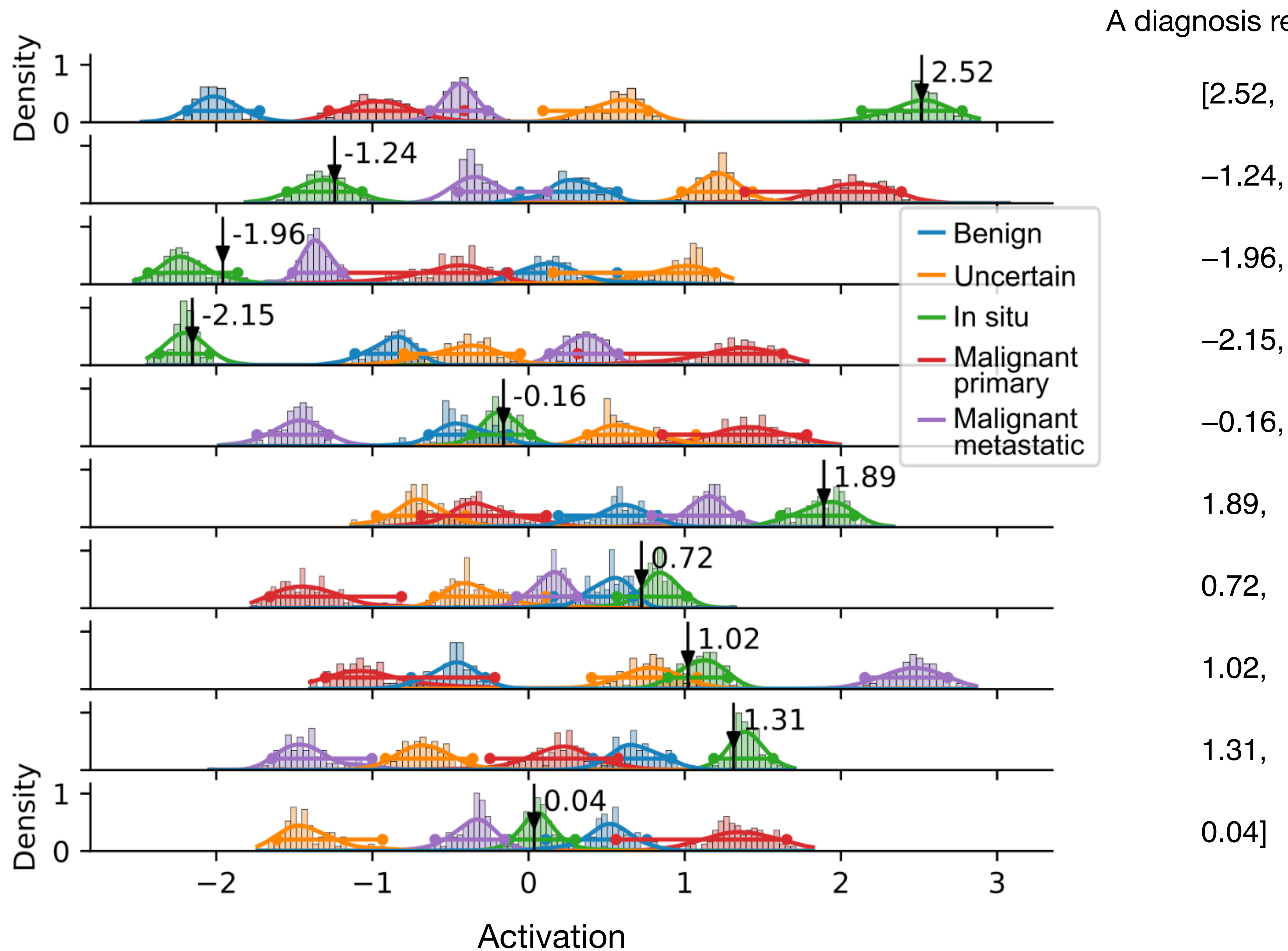
1.21,

1.06,

-1.73,

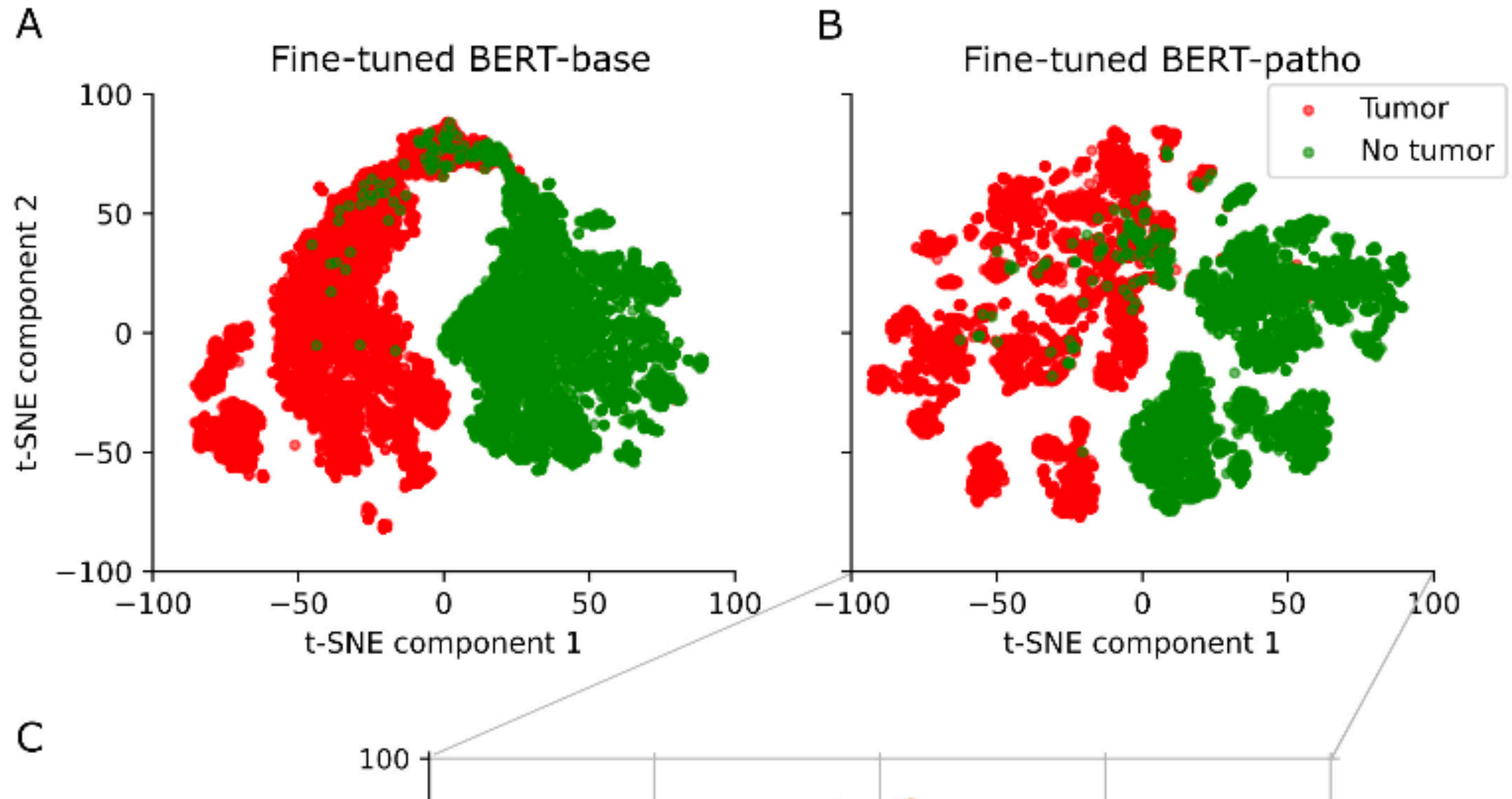
1.77]

Single dimensions of the embedding



Methods & Results - “Pathology-Knowledge” map

t-SNE dimensionality reduction of the embeddings into 2-dimensional space:
more subclusters emerge in the fine-tuned BERT-patho model (subplot B).



For finding subclusters (similar diagnoses):
DBSCAN (Density Based Spatial Clustering of Applications with Noise).

Text analysis:

- n-gram analysis of words (unigrams, bigrams, trigrams) + manual inspection
- Topic modeling [6]
- Latent Dirichlet Allocation [7]
- Evaluation of an expert pathologist

[6] George, L., & Sumathy, P. (2023). An integrated clustering and BERT framework for improved topic modeling. *International Journal of Information Technology*, 15(4), 2187-2195.

[7] Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of machine Learning research*, 3(Jan), 993-1022.

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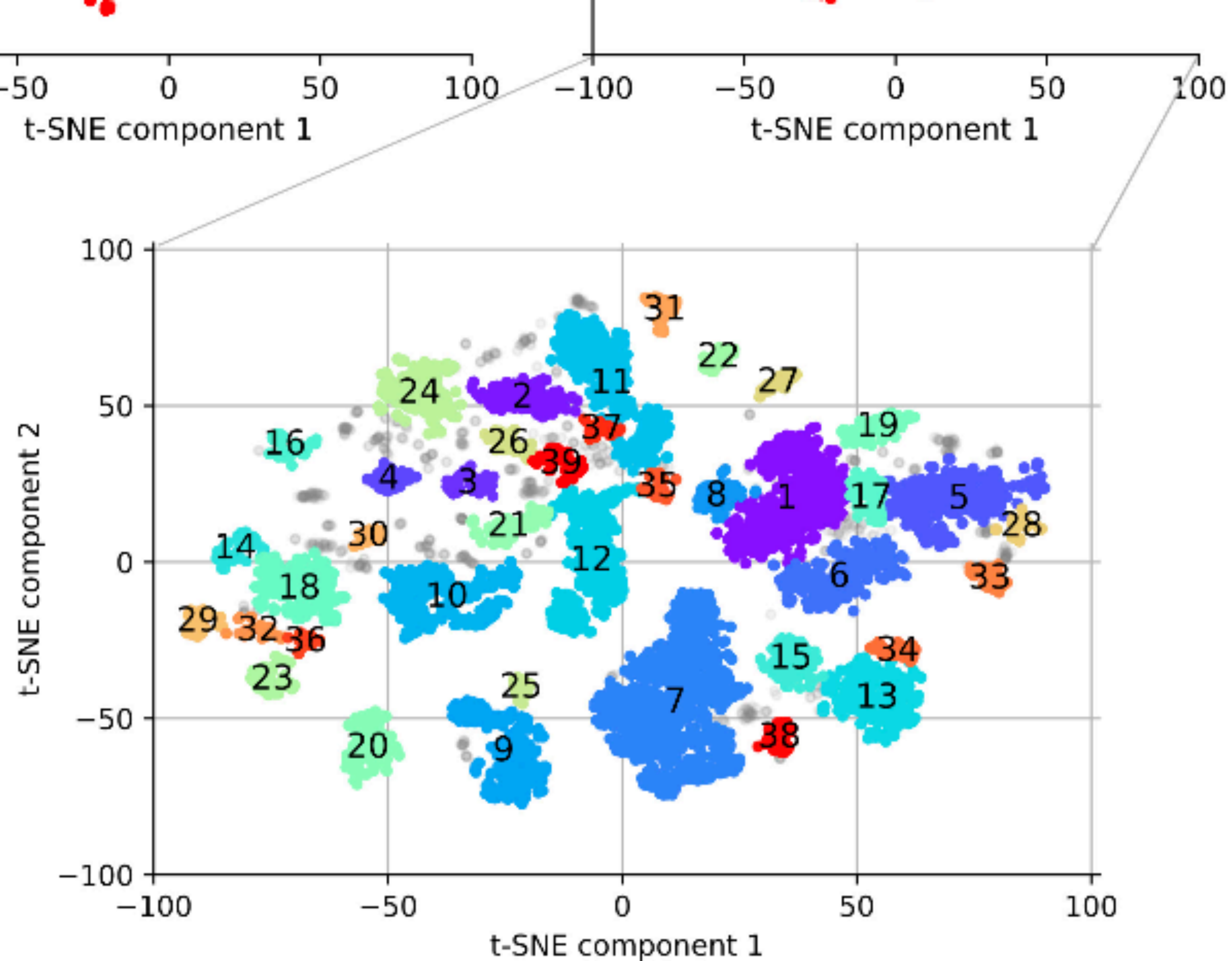


Figure 3: t-SNE of embeddings after fine-tuning on the "Tumor - No Tumor" task.

Table 9

“Tumor” cluster descriptions with English keywords translated from German

Cluster	n-gram analysis and manual inspection	Latent Dirichlet Analysis
Cluster 2	metastasis, (metastasis-free) lymph nodes	carcinoma, lymph node, differentiated, skin, metastasis, invasive, moderate, low, breast carcinoma, compatible
Cluster 3	fibroma, fibroadenoma, histiocytoma	skin, malignancy, fibroadenoma, healthy, histiocytoma, grade, dermatofibroma, parts, diagnosis
Cluster 4	basal cell carcinoma, skin piece, resection	basal cell carcinoma, healthy, solid, diagnosis, skin piece, resection, removed, tumor-free
Cluster 9	nevus cell nevus, dermal, skin, papillomatous	nevus, nevus cell, dermal, compound type, healthy, papillomatous, skin, cell, removed, malignancy
Cluster 10	adenoma, colonic mucosa, intraepithelial neoplasia, tubular	neoplasia, intraepithelial, adenoma, low-grade, colonic mucosa, parts, colonic mucosa, tubular, dysplasia
Cluster 11	breast carcinoma, (fibrocystic) mastopathy, invasive ductal, squamous epithelial carcinoma of the cervix	invasive, classification, lymph node, differentiated, breast carcinoma, low, diagnosis

Table 10

“No tumor” cluster descriptions with English keywords translated from German

Cluster	n-gram analysis and manual inspection	Latent Dirichlet Analysis
Cluster 1	(without) pathologic changes, (morphologic-) nonspecific, hyperplasia, colitis, gastric mucosa, inflammation	malignancy, chronic, indication, changes, diagnosis, present, hyperplasia, colitis, tumor tissue
Cluster 5	appendicitis, sequestering, disc degeneration, partly mucoid	appendicitis, diagnosis, disc degeneration, partial, appendix, sequestrating, ganglion, periappendicitis, acute, residuals
Cluster 6	hyperplasia, (lymphatic, chronic, lymphoid) tonsillitis, sinusitis	chronic, hyperplasia, tonsillitis, low-grade, chronic, non-specific, low-grade, residual, diagnosis, lymphadenitis
Cluster 7	chronic gastritis, gastric mucosa, antrum type	gastritis, chronic, antral type, hlo, malignancy, gastric mucosa, inactive, detectable, hyperplasia, corpus type

Conclusions

- **Linear separability** of output classes in single embedding dimensions enables visualizations
- **Clustering effects** within the embedding space:
 - Support annotation of pathology reports
 - Pathology dictionary
 - For learning purposes / training of younger pathologists
- An engineering application:
 - Use of existing machine learning methods
 - But, the workflow is novel in the pathology domain
- Limitations: relatively simple tasks, but it is also desirable to do it so (e.g., meeting explainability needs).

Thank you for listening!

Questions?

Results

Table 2

Linear Discriminant Analysis (LDA) classifier accuracy measuring how well the representations corresponding to diagnoses of pathology-specific tasks and produced by different fine-tuned models can be linearly separated. Even single dimensions of embeddings (neurons) achieve high LDA accuracy.

Task	Fine-tuned model	LDA accuracy	Single neuron accuracy [min, max, mean, median]	Proportion of neurons with accuracy ≥ 0.9
Tumor - No tumor	BERT-base	0.983	[0.403, 0.986, 0.915, 0.965]	0.755 (580/768)
	BERT-patho	0.981	[0.445, 0.982, 0.885, 0.951]	0.667 (512/768)
Tumor behavior	BERT-base	0.983	[0.360, 0.971, 0.792, 0.806]	0.145 (111/768)
	BERT-patho	0.954	[0.337, 0.943, 0.716, 0.726]	0.017 (13/768)